

Python List Methods Tasks

1. Add an Element

Problem: Create a list of 5 numbers and add a new number at the end using `append()`.

Code

```
numbers = [10, 20, 30, 40, 50]
print("Original list:", numbers)

numbers.append(60)
print("After append:", numbers)
```

Output

```
Original list: [10, 20, 30, 40, 50]
After append: [10, 20, 30, 40, 50, 60]
```

2. Add Multiple Elements

Problem: Create a list of student names and add three more names using `extend()`.

Code

```
students = ["Rahul", "Priya", "Amit"]
print("Original list:", students)

students.extend(["Sneha", "Kiran", "Divya"])
print("After extend:", students)
```

Output

```
Original list: ['Rahul', 'Priya', 'Amit']
After extend: ['Rahul', 'Priya', 'Amit', 'Sneha', 'Kiran', 'Divya']
```

3. Insert at Specific Position

Problem: Insert a new element at the beginning, middle, and end of a list using `insert()`.

Code

```
items = [10, 20, 30, 40]
print("Original list:", items)

items.insert(0, 5)          # beginning
print("Insert at beginning:", items)

items.insert(len(items) // 2, 25)  # middle
print("Insert at middle:", items)

items.insert(len(items), 50)      # end
print("Insert at end:", items)
```

Output

```
Original list: [10, 20, 30, 40]
Insert at beginning: [5, 10, 20, 30, 40]
Insert at middle: [5, 10, 25, 20, 30, 40]
Insert at end: [5, 10, 25, 20, 30, 40, 50]
```

4. Remove a Specific Element

Problem: Create a list containing duplicate values and remove only the first occurrence of a given value using `remove()`.

Code

```
numbers = [10, 20, 30, 20, 40, 20]
print("Original list:", numbers)

numbers.remove(20)
print("After removing first 20:", numbers)
```

Output

```
Original list: [10, 20, 30, 20, 40, 20]
After removing first 20: [10, 30, 20, 40, 20]
```

5. Remove Element by Position

Problem: Create a list of numbers and remove the element at a user-given index using `pop()`.

Sample input: 2

Code

```
numbers = [10, 20, 30, 40, 50]
print("Original list:", numbers)

index = int(input("Enter the index to remove: "))
removed = numbers.pop(index)

print("Removed element:", removed)
print("Updated list:", numbers)
```

Output

```
Original list: [10, 20, 30, 40, 50]
Enter the index to remove: 2
Removed element: 30
Updated list: [10, 20, 40, 50]
```

6. Remove Last Element

Problem: Remove the last element from a list using `pop()` without providing an index.

Code

```
fruits = ["apple", "banana", "mango", "orange"]
print("Original list:", fruits)

removed = fruits.pop()
print("Removed element:", removed)
print("Updated list:", fruits)
```

Output

```
Original list: ['apple', 'banana', 'mango', 'orange']
Removed element: orange
Updated list: ['apple', 'banana', 'mango']
```

7. Find Occurrence Count

Problem: Given a list with duplicate values, find how many times a particular value occurs using count().

Code

```
numbers = [1, 2, 3, 2, 4, 2, 5]
value = 2

print("List:", numbers)
print("Number of times", value, "occurs:", numbers.count(value))
```

Output

```
List: [1, 2, 3, 2, 4, 2, 5]
Number of times 2 occurs: 3
```

8. Find Position of an Element

Problem: Find the index of a given element using index().

Code

```
colors = ["red", "green", "blue", "yellow"]
element = "blue"

print("List:", colors)
print("Index of", element, "is:", colors.index(element))
```

Output

```
List: ['red', 'green', 'blue', 'yellow']
Index of blue is: 2
```

9. Sort Numbers in Ascending Order

Problem: Sort a list of numbers in ascending order using sort().

Code

```
numbers = [40, 10, 50, 30, 20]
print("Original list:", numbers)

numbers.sort()
print("Ascending order:", numbers)
```

Output

```
Original list: [40, 10, 50, 30, 20]
Ascending order: [10, 20, 30, 40, 50]
```

10. Sort Numbers in Descending Order

Problem: Sort a list of numbers in descending order using sort(reverse=True).

Code

```
numbers = [40, 10, 50, 30, 20]
print("Original list:", numbers)

numbers.sort(reverse=True)
print("Descending order:", numbers)
```

Output

```
Original list: [40, 10, 50, 30, 20]
Descending order: [50, 40, 30, 20, 10]
```

11. Reverse a List

Problem: Reverse a list without using slicing and without creating another list. Use reverse().

Code

```
numbers = [1, 2, 3, 4, 5]
print("Original list:", numbers)

numbers.reverse()
print("Reversed list:", numbers)
```

Output

```
Original list: [1, 2, 3, 4, 5]
Reversed list: [5, 4, 3, 2, 1]
```

12. Clear a List

Problem: Create a list of elements, display it, clear all elements using clear(), and display the result.

Code

```
items = ["pen", "book", "bag", "bottle"]
print("Before clear:", items)

items.clear()
print("After clear:", items)
```

Output

```
Before clear: ['pen', 'book', 'bag', 'bottle']
After clear: []
```

13. Create an Independent Copy

Problem: Create a list and make a copy using copy(). Modify the copied list and prove that the original list is unchanged.

Code

```
original = [1, 2, 3, 4, 5]
copied = original.copy()

copied.append(6)
copied[0] = 100

print("Original list:", original)
print("Copied list:", copied)
print("Same list object?", original is copied)
```

Output

```
Original list: [1, 2, 3, 4, 5]
Copied list: [100, 2, 3, 4, 5, 6]
Same list object? False
```

14. Merge Two Lists

Problem: Create two lists and combine the second list into the first using `extend()`.

Code

```
list1 = [1, 2, 3]
list2 = [4, 5, 6]
print("List 1:", list1)
print("List 2:", list2)

list1.extend(list2)
print("After merging:", list1)
```

Output

```
List 1: [1, 2, 3]
List 2: [4, 5, 6]
After merging: [1, 2, 3, 4, 5, 6]
```

15. Student Marks Processing

Problem: Given a list of student marks: add a new mark, remove a mark, sort the marks, reverse the marks. Use appropriate list methods.

Code

```
marks = [78, 92, 65, 88, 70]
print("Original marks:", marks)

marks.append(95)
print("After adding 95:", marks)

marks.remove(65)
print("After removing 65:", marks)

marks.sort()
print("After sorting:", marks)

marks.reverse()
print("After reversing:", marks)
```

Output

```
Original marks: [78, 92, 65, 88, 70]
After adding 95: [78, 92, 65, 88, 70, 95]
After removing 65: [78, 92, 88, 70, 95]
After sorting: [70, 78, 88, 92, 95]
After reversing: [95, 92, 88, 78, 70]
```

16. Shopping Cart

Problem: Create a shopping cart containing 5 items. Add an item, insert an item at position 2, remove an item, remove the last item. Use list methods.

Code

```
cart = ["milk", "bread", "eggs", "rice", "sugar"]
print("Cart:", cart)

cart.append("butter")
print("After adding butter:", cart)

cart.insert(2, "apple")
print("After inserting apple at position 2:", cart)

cart.remove("eggs")
print("After removing eggs:", cart)

cart.pop()
print("After removing the last item:", cart)
```

Output

```
Cart: ['milk', 'bread', 'eggs', 'rice', 'sugar']
After adding butter: ['milk', 'bread', 'eggs', 'rice', 'sugar', 'butter']
After inserting apple at position 2: ['milk', 'bread', 'apple', 'eggs', 'rice', 'sugar', 'butter']
After removing eggs: ['milk', 'bread', 'apple', 'rice', 'sugar', 'butter']
After removing the last item: ['milk', 'bread', 'apple', 'rice', 'sugar']
```

17. Duplicate Detection

Problem: Given a list of numbers, take a number from the user and check whether it appears more than once using count().

Sample input: 20

Code

```
numbers = [10, 20, 30, 20, 40, 50, 10]
print("List:", numbers)

num = int(input("Enter a number: "))

if numbers.count(num) > 1:
    print(num, "appears more than once.")
else:
    print(num, "does not appear more than once.")
```

Output

```
List: [10, 20, 30, 20, 40, 50, 10]
Enter a number: 20
20 appears more than once.
```

18. Move an Element

Problem: Given a list and an element, remove that element from its current position and insert it at the beginning of the list.

Code

```
numbers = [10, 20, 30, 40, 50]
element = 40
print("Original list:", numbers)

numbers.remove(element)
numbers.insert(0, element)
print("After moving", element, "to the beginning:", numbers)
```

Output

```
Original list: [10, 20, 30, 40, 50]
After moving 40 to the beginning: [40, 10, 20, 30, 50]
```

19. Second Largest Number

Problem: Given a list of numbers, use sort() and appropriate list operations to find the second largest number.

Code

```
numbers = [12, 45, 7, 89, 34, 89, 23]
print("List:", numbers)

numbers.sort(reverse=True)
largest = numbers[0]

second_largest = None
for n in numbers:
    if n != largest:
        second_largest = n
        break

print("Sorted list:", numbers)
print("Second largest number:", second_largest)
```

Output

```
List: [12, 45, 7, 89, 34, 89, 23]
Sorted list: [89, 89, 45, 34, 23, 12, 7]
Second largest number: 45
```

20. List Method Challenge

Problem: Given numbers = [40, 10, 20, 40, 30, 20, 50], perform in order: add 60, insert 5 at index 0, count how many times 20 occurs, find the index of 40, remove the first 40, remove the last element, sort the list in descending order, create a copy of the final list, clear the original list, display both lists at the end.

Code

```
numbers = [40, 10, 20, 40, 30, 20, 50]

numbers.append(60)
print("After adding 60:", numbers)

numbers.insert(0, 5)
print("After inserting 5 at index 0:", numbers)

print("Count of 20:", numbers.count(20))
print("Index of 40:", numbers.index(40))

numbers.remove(40)
print("After removing first 40:", numbers)

numbers.pop()
print("After removing last element:", numbers)

numbers.sort(reverse=True)
print("After sorting in descending order:", numbers)

copied = numbers.copy()
numbers.clear()

print("Original list:", numbers)
print("Copied list:", copied)
```

Output

```
After adding 60: [40, 10, 20, 40, 30, 20, 50, 60]
After inserting 5 at index 0: [5, 40, 10, 20, 40, 30, 20, 50, 60]
Count of 20: 2
Index of 40: 1
After removing first 40: [5, 10, 20, 40, 30, 20, 50, 60]
After removing last element: [5, 10, 20, 40, 30, 20, 50]
After sorting in descending order: [50, 40, 30, 20, 20, 10, 5]
Original list: []
Copied list: [50, 40, 30, 20, 20, 10, 5]
```